

Competitive Exam Practice 7: Microprogrammed Control Unit Design

GATE (Computer Science) Pattern

PCC CS-401: Computer Organization and Architecture

August 7, 2026

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

- Q1** [*GATE CS 2002, Q2.7, 2 marks — verified, see provenance note above*] [**MCQ, 2 marks**] Horizontal microprogramming:
- (A) does not require use of signal decoders
 - (B) results in larger sized microinstructions than vertical microprogramming
 - (C) uses one bit for each control signal
 - (D) all of the above
- Q2** [*Original, GATE-pattern*] [**MCQ, 1 mark**] Compared to horizontal microprogramming, vertical microprogramming:
- (A) requires no decoder between control memory and the datapath
 - (B) produces wider control words
 - (C) requires a decoder, adding to the delay before control signals become valid
 - (D) cannot encode mutually exclusive control signals
- Q3** [*Original, GATE-pattern*] [**NAT, 1 mark**] A set of 16 mutually exclusive control signals is to be encoded as a single vertical field. The minimum number of bits required is _____.
- Q4** [*Original, GATE-pattern*] [**MCQ, 1 mark**] In a microprogrammed control unit, the component that loads the microprogram counter (MPC) with the starting microaddress of the instruction currently in IR is the:
- (A) Control memory
 - (B) Mapping unit
 - (C) Sequence counter
 - (D) Arithmetic logic unit
- Q5** [*Original, GATE-pattern*] [**MCQ, 2 marks**] Arranged from fastest to slowest per control step, the typical ordering of the following control-unit implementations is:
- (A) Vertical microprogrammed > Horizontal microprogrammed > Hardwired
 - (B) Hardwired > Horizontal microprogrammed > Vertical microprogrammed

- (C) Horizontal microprogrammed > Hardwired > Vertical microprogrammed
(D) Hardwired > Vertical microprogrammed > Horizontal microprogrammed

Q6 [Original, GATE-pattern] [NAT, 2 marks] ADD requires 3 control-memory rows, SUB requires 3, and MUL requires 5. Assuming no rows are shared between instructions, the total number of control-memory rows needed for all three microprograms is _____.